Test 2 - Test Log

Description of test	Test data to be used (if required)	Expected outcome	Actual outcome	Comments and intended actions

Visual Debug and intial run	N/A	Code should run or provide output highlighting any syntax errors that would cause it not to run.	Line 2 – a colon ins missing in the function definition. Which will prevent it running.
			`def get_check_client_ID ()`
			`def get_check_client_ID ()`
			`^`
			`SyntaxError: expected ':'`
				Colon added.
				`def get_check_client_ID ():`

Run program to confirm fix on line 2 works	N/A	Code should progress to line 2 and highlight any other syntax errors.	Error in name when calling main functional
			`<module>`
			`man()`
			`^^^`
			`NameError: name 'man' is not defined. Did you mean: 'main'?`
			`man()`
				Name of function call corrected.
				`main()`

Run program to confirm fix on line 148 works	N/A	Code should call the main function and prompt the user to write enter their name.	Works as expected `Please enter the name of the client:`	Initial syntax errors fixed. I will move on to testing specific functionality

Check Program correctly validates the length of the client ID number	1234567890	Meets the length requirement should be accepted and prompt the user to enter amount to be entered	Works as expected
			`Please enter the name of the client: Ian`
			`Please enter the Client_ID number: 1234567890`
			`Client_ID accepted`
			`Please enter the amount to be invested per year: £`
				Code works for valid data. I will now test for how it handles invalid data

Description of test	Test data to be used (if required)	Expected outcome	Actual outcome	Comments and intended actions

Check Program correctly validates the length of the client ID number at the boundary of the logical test	123456789	Too few characters should be rejected	Did not work as expected - Number was accepted
			
			`Please enter the name of the client: Ian`
			`Please enter the Client_ID number: 123456789`
			`Client_ID accepted`
			`Please enter the amount to be invested per year: £`
				Logic of the length check get_check_client_ID function is incorrect:
				`clientID = input('Please enter the Client_ID number: ')`
				`if len(clientID) > 10 :`
				`    print('Client ID must be 10 characters long')`
				
				only checking if the length is more than 10 rather than ensuring it is exactly 10.
				
				Changed logic to != 10.
				
				`if len(clientID) != 10 :`

Check change of logic has worked.
 	
Check Program correctly validates the length of the client ID number at the boundary of the logical test
	123456789	Too few characters should be rejected	Works as expected – ID was rejected
			
			`Please enter the Client_ID number: 123456789`
			`Client ID must be 10 characters long`
			`Please enter the Client_ID number:`
				Confirm logic works by testing upper boundary

Check change of logic has worked.
 	
Check Program correctly validates the length of the client ID number at the boundary of the logical test
	12345678910	Too many characters should be rejected	Works as expected – ID was rejected
			
			`Please enter the Client_ID number: 12345678910`
			`Client ID must be 10 characters long`
			`Please enter the Client_ID number:`
				Issue resolved

Check standard savings plan option provide correct outputs	Amount invested per year = 1000	Maximum value = 4245.82
		
		Minimum value = 4121.44
			Program crashes before the calculations occur:
			
			`investment_type`
			`    if choice == "1" and investmentAmount >= 20000:`
			`                         ^^^^^^^^^^^^^^^^^^^^^^^^`
			`TypeError: '>=' not supported between instances of 'str' and 'int'`
				get_chek_investment_amount function is returning the value as a string instead of as a numerical data type.
				`return amount`
				
				Changed to return the value as a float data type.

<table>
<tr><th>Description of test</th><th>Test data to be used (if required)</th><th>Expected outcome</th><th>Actual outcome</th><th>Comments and intended actions</th></tr>
<tr>
<td>Check standard savings plan option provide correct outputs</td>
<td>Amount invested per year = 1000</td>
<td>Maximum value = 5371.71
Fees for max = 39.67
Minimum value = 5182.89
Fees for min = 38.55</td>
<td>Maximum and minimum value are incorrect
Fees are also incorrect.
<pre>
********** Investment summary **********
Client name: Bob
Client ID: 1234567890
****
********* Savings plan summary *********
Initial Investment: £ 1000.0

If the savings plan perfoms at the highest rate after 5 years you can expect:
Value of investment: £ 4245.83
Fees paid: £ 26.22
Profit made: £ -780.39

If the savings plan perfoms at the lowest rate after 5 years you can expect:
Value of investment: £ 4121.45
Fees paid: £ 25.61
Profit made: £ -904.16
</pre></td>
<td>Maximum and minimum should be at least 5000. As it is 1000 per year. This suggests that it is not calculating the correct number of years.

The range that controls the for loop is incorrect. A range of 1 to 5 would only loop 4 times.
<pre>
def savings_plan (investmentAmount):
    maximumValue = 0
    minimumValue = 0
    feesMax = 0
    feesMin = 0

    for i in range (1,5):
</pre>
Changed the lower number to 0 as counting for loops and indexes starts at 0.
<pre>
    for i in range (0,5):
</pre></td>
</tr>
<tr>
<td>Check updated loop corrects the calculation

Check standard savings plan option provide correct outputs</td>
<td>Amount invested per year = 1000</td>
<td>Maximum value = 5371.71
Fees for max = 39.67
Minimum value = 5182.89
Fees for min = 38.55</td>
<td>Calculations re now correct.
<pre>
********** Investment summary **********
Client name: Jane
Client ID: 1234567890
****
********* Savings plan summary *********
Initial Investment: £ 1000.0

If the savings plan perfoms at the highest rate after 5 years you can expect:
Value of investment: £ 5371.73
Fees paid: £ 39.65
Profit made: £ 332.08

If the savings plan perfoms at the lowest rate after 5 years you can expect:
Value of investment: £ 5182.91
Fees paid: £ 38.56
Profit made: £ 144.34
</pre>
Note – There is a slight difference between my calculated values and the program output, but this is because I rounded my calculations to make them easier.</td>
<td>Basic savings option tested and issues resolved.</td>
</tr>
</table>

Description of test	Test data to be used (if required)	Expected outcome	Actual outcome	Comments and intended actions

Check that basic savings plan limits investment to 20000 per year	Amount per year = 20000	Value accepted.	It did not accept he value of 20000 for the basic savings
			
			`Enter choice here: 1`
			`The amount you wish to invest is too high for the standard savings plan.`
			`You will be shown an example for managed stock investments instead`
			`********** Investment summary **********`
				The logic is incrrect in the investment type funciton.
				
				`investmentAmount >= 20000:`
				
				Should be > 20000 not >=20000
				
				`investmentAmount > 20000:`

Check issue has been resolved	Amount per year = 20000	Value accepted.	This now accepts the 20000 for a basic investment.
			
			And calculates correctly.
			
			`********* Savings plan summary *********`
			`Initial Investment: £ 20000.0`
			
			`If the savings plan perfoms at the highest rate after 5 years you can expect:`
			`Value of investment: £ 107434.59`
			`Fees paid: £ 793.02`
			`Profit made: £ 6641.56`
			
			`If the savings plan perfoms at the lowest rate after 5 years you can expect:`
			`Value of investment: £ 103658.12`
			`Fees paid: £ 771.25`
			`Profit made: £ 2886.87`	

Check managed stocks calculations	Amount per year = 1000	Maximum value – approx 9700
		
		Min value – approx 5600
			Upper calculation is correct, but the minimum value is higher than the maximum,
			
			`** Managed Stocks Investment Summary ***`
			`Initial Investment: £ 1000.0`
			
			`If the managed stocks plan perfoms at the highest rate after 5 years you can exp`
			`Value of investment: £ 9707.9390943`
			`Fees paid: £ 327.3041135559`
			`Profit made: £ 4380.6349807441`
			
			`If the managed stocks plan perfoms at the lowest rate after 5 years you can expe`
			`Value of investment: £ 15323.84`
			`Fees paid: £ 469.7347199999999`
			`Profit made: £ 9854.10528`
				Minimum value calultion is uing 40% instead of 4%
				
				`minimumValue = minimumValue + investmentAmount`
				`minimumValue = minimumValue + (minimumValue * 0.40)`
				
				Changed:
				
				`minimumValue = minimumValue + investmentAmount`
				`minimumValue = minimumValue + (minimumValue * 0.04)`

Description of test	Test data to be used (if required)	Expected outcome	Actual outcome	Comments and intended actions
Rerun code to check calculaiton is now correct	Amount per year = 1000	Maximum value – approx 9700 Min value – approx 5600	Functions as expected ** Managed Stocks Investment Summary *** Initial Investment: £ 1000.0 If the managed stocks plan perfoms at the highest rate after 5 years you can expect: Value of investment: £ 9707.9390943 Fees paid: £ 327.3041135559 Profit made: £ 4380.6349807441 If the managed stocks plan perfoms at the lowest rate after 5 years you can expect: Value of investment: £ 5632.9754624 Fees paid: £ 213.94570629119997 Profit made: £ 419.02975610880003	No further action on this issue required

Add more rows and tables as required